Leah Do
LV Onboarding Project

In-Helmet Audio Notes

Speaker driver- component in speaker system that convert electrical signals into sound waves

- Current flows through coils, creating a small magnetic field
- Coil's magnetic field pushes and pulls against the magnet in the speaker, the coil and the diaphragm cone connected to it to move along with push and pull force created by the magnetic field
- Diaphragm pushes and pulls air, creating sound waves

Components of in helmet audio- speaker driver, dac/adc, amplifier, microphone, connectors

https://www.aperionaudio.com/blogs/aperion-audio-blog/which-is-better-4-ohm-or-8-ohm-speakers?srsltid=AU7gw4UtLM2zDMMLanyznzTyCc4ACz5SB-HRf3gK93Ncx4NkXWG9mzub

Impedance:
How much audio component, how much current they draw from the amplifier
**match input impedance to output impedance
**amplifier needs to match the nominal speaker impedance → overloading + damage
→ 8 ohms vs 4 ohms → lower impedance means less resistance, drawing more current from the amplifier gives the same amount of voltage. This means a lower impedance speaker is more prone to overloading/overheating the amp
→ higher impedance has more resistance, operates at more stable and cooler condition
→ CONCLUSION: 8 ohm impedance preferred

Pros + Cons of Larger Speaker Drivers:
cons:

- may not fit as well into the ear pocket
- may not be as comfortable to wear either
- needs more power to move heavier cones in module

pros:

- Higher volume
- Stronger bass

pros of smaller drivers: (10-20mm)

- Ideal if focusing on mid range frequency
- requires less power

→ deduced that for reproducing simple mid-range vocals and best comfort and fit, at the cost of weaker bass and lower volume, it is ideal to use a smaller speaker driver

Sound Pressure Level (SPL) = how loud the speaker will play measured typically with 1W of input heard from 1 meter away, (in dB or dBA)

dBA = A-rated decibels that filter out high and low frequencies to be more accurate to what the human ear will hear

To compare: Apple iPhone 13 built-in speaker SPL lies at around 104.7dBA

GOAL: Find one w/ 8 ohm impedance

Option 1: → NO bc dimensions too big
https://www.digikey.com/en/products/detail/pui-audio-inc/AS05008MS-R/1745550
- SPL: 85 ± 3 dBA at 1W/1M)
- Power Rating: 1.5 W & 3W max
- Dimensions: (50.50mm x 50.50mm)
- Height: 12 mm
- Frequency Range: 800 Hz - 7k Hz
- has solder eyelets for wires

Option 2: → NO bc spring type termination → not easily mountable on pcb
https://www.digikey.com/en/products/detail/soberton-inc/SP-1511S-1/6099099
- SPL: 83±3dB at 0.1m
- Power Rating: 700 mW & 800 mW max
- dimensions: (15 mm x 11 mm)
- height: 3.15 mm
- frequency rating: 400 Hz ~5k Hz
- Spring type termination: solderless and mechanical spring fingers need to press down onto pcb → must add custom surface mount pads for that

Option 3: (CHOOSING THIS ONE)
Website: https://www.digikey.com/en/products/detail/pui-audio-inc/AS02708CO-R/1745517
Datasheet: https://puiaudio.com/file/specs-AS03208MS-3-R.pdf
- SPL: 85 ± 3 dB at 0.1W/0.5m
- Power Rating: 1 W
- Dimensions: 1.063" L x 0.787" W (27.00mm x 20.00mm)
- Height:0.236" (6.00mm)
- Frequency range: 250 Hz ~ 20 kHz
- termination type: solder eyelets

Would this require a DAC for the speaker driver and ADC for mic?

- DAC needed to pass the digital information into a continuous analog electrical wave that the speaker driver needs to move cones
- also need amplifier to boost electrical current so speaker driver has enough power to push the speaker cone of the driver
- amplified analog wave then travels down the wire to speaker driver in ear pocket to vibrate its cone the create to hear the persons voice

CONNECTORS:

Molex Connectors: has clip mechanism for quick connection and disconnection (4 POS, 3mm)
Plug: https://www.digikey.com/en/products/detail/molex/0430250400/252497
Receptacle: https://www.digikey.com/en/products/detail/molex/0430200401/323611
→ NO; not too good for breakaway when driver runs out of car abruptly

4mm magnetic connector (can solder/crimp 4 wires)
web:https://www.digikey.com/en/products/detail/mill-max-manufacturing-corp/878-20-004-00-012000/22237186?gclsrc=aw.ds&gad_source=1&gad_campaignid=17336967819&gbraid=0AAAADrbLIjHM4wVUXdKTL_ohOJxKqsCS&gclid=Cj0KCQjwuJLWBhD_ARIsAIBcRUxTArQdrEE_0goYh6lzJhuSulfYsa45-Y2ocsWhOUogZo7g5ByLRP0aAtJrEALw_wcB
web:https://www.digikey.com/en/products/detail/mill-max-manufacturing-corp/879-22-004-00-011101/22237181
→ NO; connection not secure enough

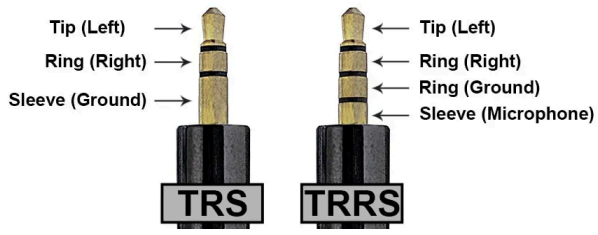
Audio Jack connectors: (THIS ONE)

3.5 mm STEREO

- FEMALE: https://www.digikey.com/en/products/detail/same-sky-formerly-cui-devices/SJ5-43502PM/1956558
- MALE: https://www.digikey.com/en/products/detail/amphenol-nexus-technologies/TP-120/9687780

Connector Jack Types:

TRS Jacks - needed for connecting two speaker drivers
TRRS Jacks - to connect speaker drivers and mic in one plug
→ **USE TRRS** if you want to allow speaker output while simultaneously receiving voice input from mic



Microphone:
MEMS I2S Digital mic with adc and amp already built into chip (THIS ONE)

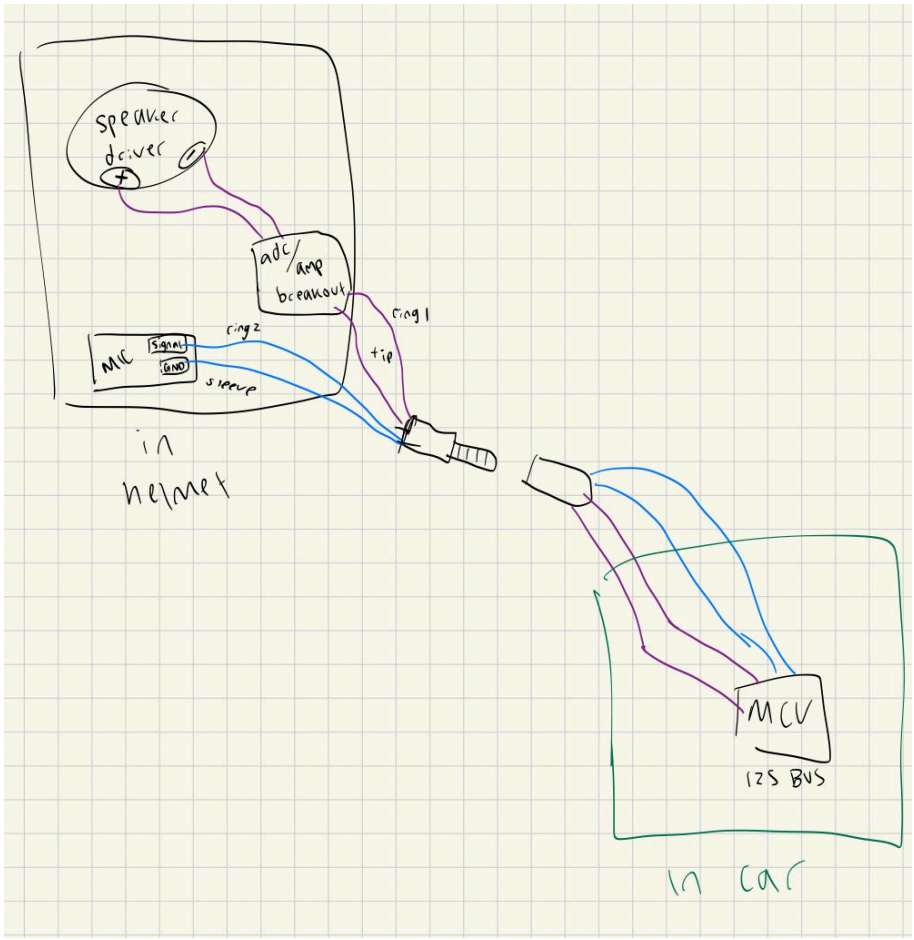
- Website Link: https://www.digikey.com/en/products/detail/tdk-invensense/MMICT5848-00-012/18634578
- Datasheet: https://mm.digikey.com/Volume0/opasdata/d220001/medias/docus/8902/MMMICT5848-00-012.pdf
- need to connect to breakout board, can't solder wires directly b/c super tiny
- Breakout in helmet??

→ add mic windscreen covers to reduce external noises like from wind?

**needs to interface with mcu thru i2s digital bus
→ **What is I2S?**
- digital sound communication protocol for transmitting audio data between devices and MCUs
- requires devices to be connected on bus with 3 wires:
    serial clock (SCK): syncs transmission of audio data
    word select (WS): switches b/w left and right audio information(ex:1 for right and 0 for left)
    serial data (SD): carries the binary audio data

Next steps…KiCAD a schematic for amp/adc breakout for mic
Adafruit Stereo I2S 3W Amplifier Breakout - Dual MAX98357A → reference for schematic
** built-in I2S DAC and Amplifier
https://www.adafruit.com/product/6513?srsltid=AU7gw4WYAdYIf4XgLNHUmhONIASu7OVqloRnMazVzXgbJQo7gmEYjCCJZ20

Uses 2 MAX98357A digital audio amplifier chips (one for left speaker and one for right speaker)
→ for my schematic, only incorporate one speaker (mono)
Data Sheet:
https://www.analog.com/media/en/technical-documentation/data-sheets/MAX98358.pdf



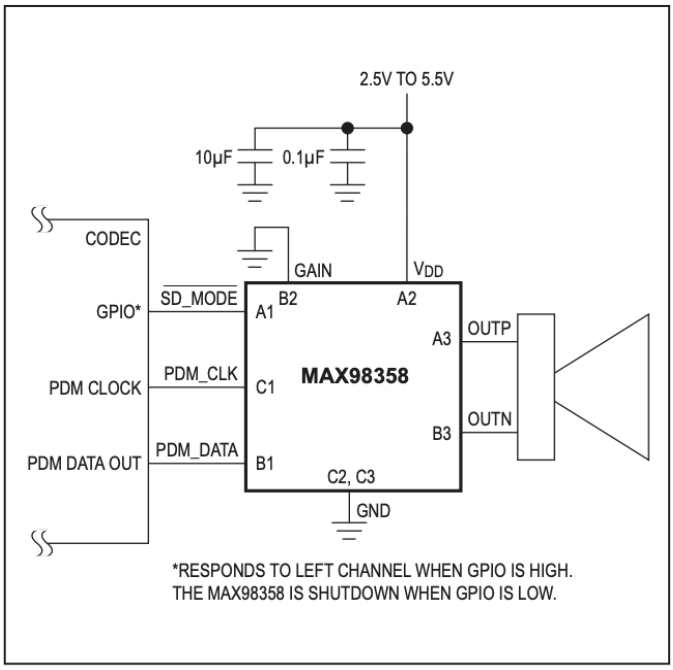


Figure 7. Left-Channel Operation with 12dB Gain

BILL OF MATERIALS

Component	Units	link	price
Speaker Driver	1	https://www.digikey.com/en/products/detail/pui-audio-inc/AS02708CO-R/1745517	$5.26
Audio Amp/ADC IC	1	https://www.digikey.com/en/products/detail/analog-devices-inc-maxim-integrated/MAX98358ETE-T/5050233\	$4.10
Audio TRRS Jack (Male)	1	https://www.digikey.com/en/products/detail/amphenol-nexus-technologies/TP-120/9687780	$18.75 → find cheaper one
Audio TRRS Jack (Female)	1	https://www.digikey.com/en/products/detail/same-sky-formerly-cui-devices/SJ5-43502PM/1956558	$3.32
Microphone	1	https://www.digikey.com/en/products/detail/tdk-invensense/MMICT5848-00-012/18634578	$2.88
		TOTAL	$34.31